

THE PROBLEM

**AQI tells cities how bad the air is.
It does not tell them what action the evidence justifies.**

WHAT THE DASHBOARD SAYS

Severe

1288 $\mu\text{g}/\text{m}^3$ PM2.5



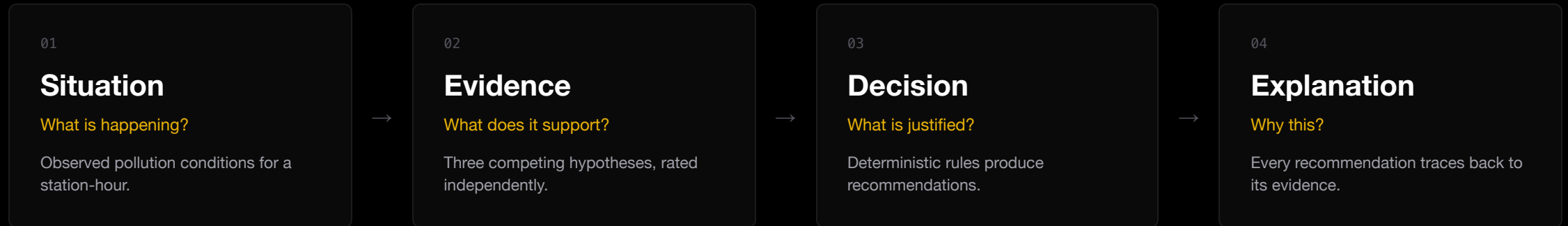
WHAT THE OFFICER STILL HAS TO ANSWER

- Why is this happening?
- What evidence do we actually have?
- What action is justified right now?
- How confident should I be?
- Can I defend this decision tomorrow?

Cities have the monitoring data. What is missing is the reasoning layer between a reading and a defensible decision, and officers assemble it by hand.

THE SOLUTION

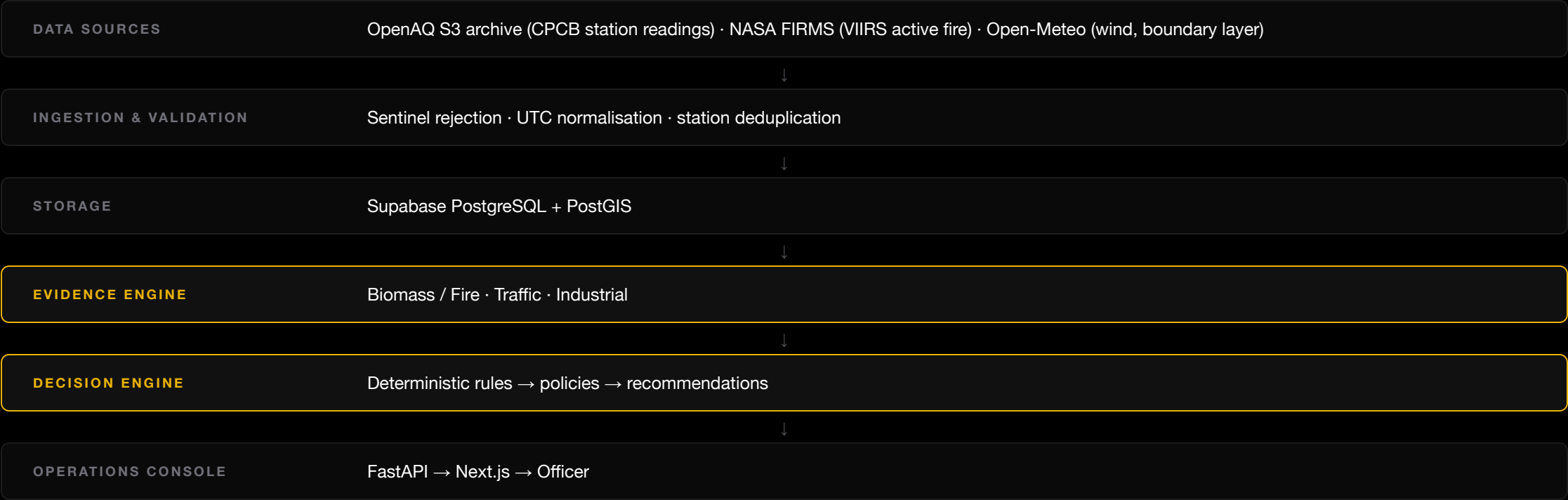
Vayu Console turns environmental evidence into transparent operational decisions.



The layout of the console is this sequence, left to right. The interface is the workflow.

HOW IT WORKS

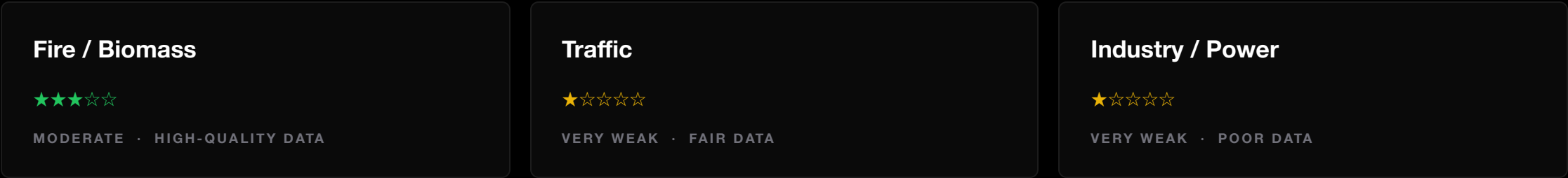
Architecture



Three ingested sources. No LLM anywhere in the decision path.

THE EVIDENCE ENGINE

We evaluate hypotheses. We do not fabricate attribution.



Actual output for the Diwali 2019 worked example.

EACH HYPOTHESIS EXPOSES

- | | | |
|---------------------------|------------------------------|-------------------------|
| · Supporting observations | · Contradicting observations | · Assumptions |
| · Evidence quality | · Limitations | · Historical validation |

Evidence strengths are independent. They are not probabilities, they do not sum to 100%, and there is no pie chart anywhere in the product.

SCIENTIFIC HONESTY

**Our first source classifier hit 98.6% accuracy.
We deleted it.**



The model had learned the labelling rule, not the atmosphere.

No labelled dataset of pollution source exists for Delhi, because producing one would itself require apportionment. The only labels available were ones we invented from a rule over our own features. Training on that reproduces the rule.

13,685 real station-hours · 5-fold CV · accuracy 0.9859 ± 0.0035 · macro F1 0.9851

WE REJECTED

Fabricated source accuracy

WE BUILT

Evidence-based hypothesis screening

Statistical machinery cannot manufacture ground truth that does not exist.

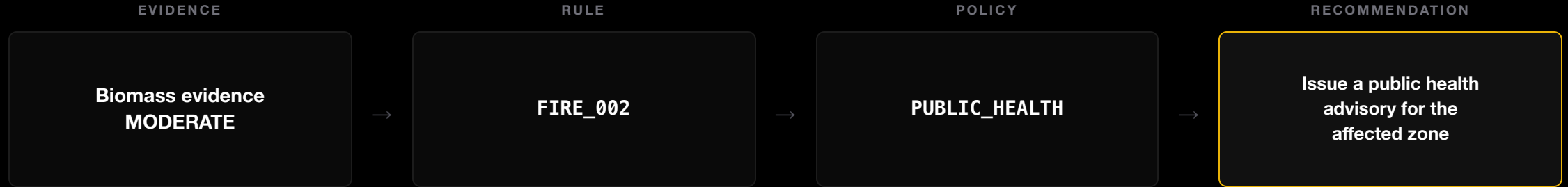
HISTORICAL VALIDATION

Each module declares the intervention that would reject it.

COVID lockdown 2020 TRAFFIC MODULE	ACCEPTED 47 stations · 901,160 rows. NO₂ -54.4%, SO₂ -3.7%. NO₂/SO₂ ratio 2.82 → 1.34. Likelihood ratio 2.11 (weak).
Diwali 2019 FIRE DISCRIMINANT VALIDITY	PENDING 1,604 VIIRS detections that day, so the confound is real. VIIRS overpasses Delhi 12:00–14:00 and 01:00–03:00 IST, outside the firework window. Structural argument, not yet a result.
Odd-Even II, Apr 2016 VEHICULAR	DATA INGESTED — VALIDATION PENDING The only unconfounded vehicle window in the record: no stubble, no winter inversion. Weak treatment on few stations.
One test has been run to completion. The engine reports the rest as pending at /evidence/history, and the console shows them as pending.	

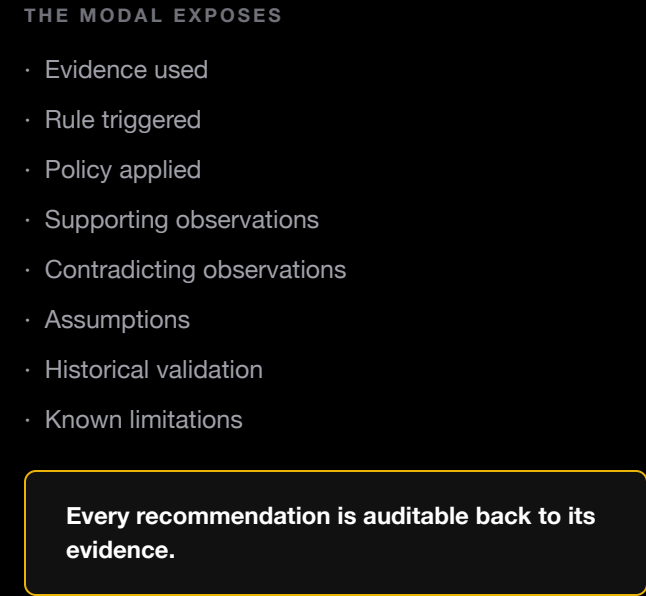
FROM EVIDENCE TO ACTION

Every recommendation has to earn its way in.



- No LLM decides enforcement actions.
- The decision engine is deterministic: the same evidence returns the same recommendation, every time.
- A rule fires on the presence of evidence, never on its absence.

Do not trust the recommendation. Challenge it.



Every recommendation is auditable back to its evidence.

Historical Replay – Diwali 2019

READING ORDER

- 1 Situation
- 2 Transport geometry
- 3 Evidence
- 4 Recommendations
- 5 Challenge
- 6 Historical validation

Every scenario is a reconstruction from archived observations. The console says so on every screen. Nothing here is a live feed.

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TECHNICAL DIFFERENTIATION

Built for evidence integrity, not just model output.

01	Resilient data foundation	OpenAQ's live API advertises Delhi stations it will not return history for. Coverage was probed against the raw S3 archive instead, the only source whose claimed coverage matches what it returns.
02	Data quality at the boundary	Missing-data sentinels (−999, −9999, any negative concentration) are dropped at parse time, not filtered in queries, so one can never reach the database and be forgotten by a later aggregate.
03	Station identity	One physical station appears under several OpenAQ location ids. Left unresolved this overstates the network and double-counts readings into city means.
04	Observation gaps	A satellite non-observation is not a true non-detection. VIIRS passes over Delhi twice a day, so an absence of fire detections may be an absence of overpass, and the system reports it that way.
05	Auditable decisions	Evidence → Rule → Policy → Recommendation, exposed in the interface rather than buried in a log.

IMPACT & FUTURE

From a Delhi proof of concept to city-scale environmental decision support.

BUILT AND DEPLOYED — DELHI

- Evidence Engine, three hypotheses
- Deterministic Decision Engine
- Operations Console with full audit trace
- COVID lockdown validation, completed

NOT YET BUILT

- Road-density context for the traffic hypothesis
- Industrial proximity for the industrial hypothesis
- Expanded historical validation (Diwali, Odd-Even)
- Live ingestion scheduling
- Additional cities, each needing its own calibration
- Policy-specific integrations

The architecture ports to another city. The calibration does not — every likelihood ratio is measured against a local natural experiment, so scaling is a data problem, not a config change.

Most systems are designed to always give an answer.

**Vayu Console is designed to give an answer only when the evidence can support one,
and to show exactly why.**

Vayu Console

Team: singhpranav431

ET AI Hackathon 2.0

Console vayu-console-web.vercel.app/console

API vayu-console-api.onrender.com

Code github.com/Pranavsingh431/vayu_console